

Step 1 - Technical Design: Multi-Source Candidate Data Transformer

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1. Pipeline / Step Breakdown

- Detect & Load: Read input files (CSV, JSON, PDF) and identify the source type.
- Extract:
 - Structured: Map fields from CSV/JSON to the canonical schema.
 - Unstructured: Use NLP/Regex to parse free-text (resume PDFs) into structured fields.
- Normalize: Standardize individual fields (e.g., dates to YYYY-MM, phones to E.164, countries to ISO-3166).
- Merge & Confidence: Group records by candidate using email/name. Resolve conflicts using source-priority and calculate an overall confidence score.
- Project-to-Output: Apply the runtime JSON configuration to reshape the record into the desired output.
- Validate: Ensure the final projected output conforms to the requested schema.

2. Canonical Output Schema & Normalized Formats

- Dates: YYYY-MM using dateutil.
- Phones: E.164 (e.g., +14155551234) using phonenumbers.
- Country: ISO-3166 alpha-2.
- Skills: Lowercased, stripped, mapped to a canonical list.
- Representation: Pydantic models for strict typing.

3. Merge / Conflict-Resolution Policy & Confidence

- Match Keys: Primary key is email. Fallback is full_name.
- Conflict Resolution: Source priority: 1) ATS/Structured Data (Highest), 2) Direct APIs, 3) Resume Parsing, 4) Recruiter Notes (Lowest).
- Confidence Scoring: ATS (0.95), API (0.90), Resume (0.80), Notes (0.60).
- Provenance: Tracks {"field": "...", "source": "...", "method": "..."} for every populated field.

4. Runtime Custom-Output Config (Projection)

- Iterates over config.fields. Uses custom traversal to fetch values from the internal canonical record using the "from" path (e.g., emails[0]).
- Applies field-level normalize rules.
- Handles missing values based on "on_missing" (null, omit, error).
- Includes or excludes provenance and overall_confidence based on flags.

5. Edge Cases & Handling

- Garbage/Missing Source File: Pipeline logs an error for the specific file but continues processing other files (Robustness).
- Un-normalizable Phone: Kept as-is with lower confidence or set to null based on strictness. Never crashes.
- Missing Match Key: Treat as a new candidate to avoid incorrect merges.
- Time-pressure Descoping: Mocked NLP extraction with RegEx heuristics to avoid external LLM dependencies for the assignment.